

HELIOLITE · DESIGN DECISIONS REGISTER

HelioLite design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-02	Amish Chadha	Decisions of 2026-10-02 carried into the model, BOM and calculations: value engineering at USD 499; R13 margin now 0.01 kg
0.3	2026-10-02	Amish Chadha	Design for construction (HLT-DDR-003, Table 1, P1 to P14) accepted by Amish on 2026-10-02; added to decisions made
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations for all open decisions 1 to 11 on 2026-10-02 (aluminum latch bracket, two-lift head fitting, edge guard, clear lid, glare check before uncovering); moved to decisions made; value engineering savings updated
0.1	2026-10-01	Amish Chadha	Register opened with the constructable design (HLT-DDR-003); budget treated as a value-engineering target

HelioLite design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The NMRV030 gearbox's output-face hole pattern (the model assumes four M6 on a 44 mm square), its mass (1.2 kg assumed) and its listed backlash	Sets the holes in the right arm and cap plate; R13 has a 0.01 kg margin with the aluminum latch bracket and the edge channel decided on 2026-10-02; R4 depends on backlash	HLT-DDR-003; HLT-CAL-001
2	A 14 mm double output shaft is sold for the chosen azimuth gearbox, and the gearbox's permissible radial load and overturning moment on its output	The yoke's wind moment passes through the azimuth gearbox's output bearings	HLT-DDR-003
3	The floor flange's hole circle (105 mm assumed) and that the pipe thread engages fully	Sets the tapped holes in the cap plate	HLT-DDR-003
4	The keyed flange hub fits the 14 mm keyed shaft, has a 30 mm boss and a 40 mm screw circle	Sets the turntable disc's holes	HLT-DDR-003
5	The release solenoid's stroke (12 mm needed), pull force with its return spring, and mass	The pawl must clear the lug fully	HLT-DDR-003
6	The spiral springs give about 3 N·m over the axis travel and fit the 50 mm and 52.5 mm cans	The preload that R4 depends on	HLT-DDR-002, N2
7	The ground screw socket fits 60.3 mm pipe and its overturning rating is at least 1.0 kN·m	Mast survival at 35 m/s	HLT-CAL-001 [E4]
8	A published stowed force and hinge moment coefficient for small heliostats	The latch is designed on an assumed coefficient with a factor of 2	HLT-CAL-001 [E7]
9	The polyurethane pad's modulus (50 MPa assumed)	The pad must deflect less than the worm's free travel	HLT-CAL-001 [E8]
10	The controller box's mounting lugs	Sets the plate's box holes	HLT-DDR-003

■ Value engineering

Value-engineering target: USD 455 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 499 (USD 44 over the target). Main cost drivers and savings worth trying:

- The largest lines are the two drives (azimuth with cap and flange USD 68, elevation USD 52), the controller with its mounting plate (USD 51), the gimbal yoke (USD 38), the backing panel, ribs, tube and trunnions (USD 36), the mast and the ground anchor (USD 35 each), the glass mirror with its edge channel (USD 31), and the turntable stack (USD 28).
- Making the design constructable repriced lines 2, 3, 5, 6, 9, 13, 14 and 17, from USD 451 to USD 492: the floor flange and tapped cap (USD 8), the shaft, hub and thrust washer (USD 13), the controller plate and U-bolts (USD 6), the anemometer clamp (USD 4), more fixings (USD 4), the rib, end block and stub parts (USD 6) and the latch detail (USD 2), partly offset by the aluminum yoke costing less than the printed one (USD 2 less). The decisions of 2026-10-02 add USD 7, to USD 499: the EPDM edge channel (line 1, about USD 6) and the aluminum latch bracket (line 17, about USD 1); the clear-lid box (line 9) lists at about the same price.
- Savings worth trying: buy both gearboxes and motors as one kit (often USD 10 to 15 less than two singles); print the turntable disc in ASA instead of aluminum if a stiffness check allows (about USD 6); a forecast-only storm stow where the forecast is reliable, making the anemometer optional (USD 24, but it weakens R9's backup). The cheaper box without a clear lid is no longer a saving: the clear lid was decided on 2026-10-02.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D10: 60.3 mm galvanized steel pipe mast; forecast plus cup anemometer; supercapacitor stow reserve; budget \$430; pitch "with no sun sensors"; face-down stow; calibration by jogged spot points; Grena algorithm 5 with weekly network time; listed indoor 12 V adapter; glass mirror with safety backing film	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	HLT-DDR-001
2026-09-25	TRL 3 review items N1 to N6: stow stop and latch carrying the stowed moment; spiral preload springs on both axes; R13 relaxed to 13 kg; R7 and R10 reworded; four calibration points over about 4 h	Amish: "i accept all your recommendations, go with them across all repos."	HLT-DDR-002
2026-09-26	Budget top-up to \$455 for the \$451 BOM (P1, option a)	Amish: "I am ok with the budget top ups"	HLT-DDR-002 v0.2
2026-09-30	Make the design physically buildable while drawing the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	HLT-DDR-003 (its Table 1 changes and its Table 3 items were decided on 2026-10-02, below)
2026-10-01	budget_usd is a hypothetical value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	This register, Value engineering
2026-10-02	First site and user: a house first, owned by a contributor or someone in the team's network, chosen by the siting check of decision 3 (one mirror spot sees both the winter sun and a north window); schools only after glare safety is shown	Amish: "i approve your recommendations for all 555 open decisions."	HLT-DDR-001, 01
2026-10-02	Local rules and glare: before the mirror is uncovered, check the local building and planning rules and do a simple glare study of where the beam can go (neighbours, roads, sky); the beam stays inside the property in every state	Amish: "i approve your recommendations for all 555 open decisions."	HLT-DDR-001, 02
2026-10-02	Siting survey: a desk survey of 20 to 30 real yards using satellite images and a sun-path check, scored on whether one mirror spot sees the winter sun and a north window within range, then a visit to the first build site	Amish: "i approve your recommendations for all 555 open decisions."	HLT-DDR-001, 03
2026-10-02			

Date	Decision	Decided by	Record
	Benefit: public copy leads with daylight and treats heat as a bonus until co-design shows otherwise; both questions are asked at the co-design sessions	Amish: "i approve your recommendations for all 555 open decisions."	HLT-PRC-001, open questions
2026-10-02	R13 mass margin (option b, changed from a): the latch bracket is made in aluminum now (about 0.1 kg saved), and the parts are weighed at TRL 4	Amish: "i approve your recommendations for all 555 open decisions."	HLT-DDR-003, A1
2026-10-02	Head on the mast (option b, changed from a): for the prototype the yoke and drive are fitted at height first and the mirror assembly (about 6.2 kg) is lowered into it, glass covered, from a stable platform rather than steps; a hinged mast base (option c) is looked at before any second build	Amish: "i approve your recommendations for all 555 open decisions."	HLT-DDR-003, A2
2026-10-02	Appearance model and renders: updated to the aluminum yoke, the new latch and the rib tunnels, dropping the torque tube saddles	Amish: "i approve your recommendations for all 555 open decisions."	HLT-DDR-003, A3; docs/REVIEW.md , 2026-09-26, item 4
2026-10-02	Target distance in the product render: the wall stays 1.0 m from the mast as a render-only layout, with the caption saying so	Amish: "i approve your recommendations for all 555 open decisions."	docs/REVIEW.md , 2026-09-26, item 1
2026-10-02	Mirror edge guard: a black EPDM edge channel round the glass and panel edges is added to the BOM, paired with the aluminum latch bracket of decision 5; if the weighed head still exceeds 13 kg, R13 is relaxed slightly rather than the guard dropped	Amish: "i approve your recommendations for all 555 open decisions."	docs/REVIEW.md , 2026-09-26, item 2
2026-10-02	Controller lid: an IP65 box with a clear lid is specified on BOM line 9	Amish: "i approve your recommendations for all 555 open decisions."	docs/REVIEW.md , 2026-09-26, item 3
2026-10-02	Product finish set (option a), with the yoke in natural aluminum: teal turntable and name plate, silver-grey gearboxes, black motors, galvanized mast and cap, dark anchor	Amish: "i approve your recommendations for all 555 open decisions."	docs/REVIEW.md , 2026-09-26, item 5
2026-10-02	Design for construction accepted: the changes P1 to P14 in Table 1, as made	Amish: "APPROVED: Design-for-construction changes in 10 repos (CityTwin, CoolShade, PalletPilot, Heliolite, PotholeLog, EarthPress, ReadyKit, CellCheck, CargoMule and ThermaCart)"	HLT-DDR-003, Table 1